CBSE Test Paper 03

Chapter 9 Mechanical Properties of Solids

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1. In constructing a large mobile, an artist hangs an aluminum sphere of mass 6.0 kg from a vertical steel wire 0.50 m long and 2.5 $\times$ 10^{-3} cm^2 in cross-sectional area. On the bottom of the sphere he attaches a similar steel wire, from which he hangs a brass cube of mass 10.0 kg. Compute the elongation. **1**
   a. 1.5 mm upper, 1.1 mm lower
   b. 1.4 mm upper, 1.0 mm lower
   c. 1.6 mm upper, 1.0 mm lower
   d. 1.7 mm upper, 1.0 mm lower

2. A circular steel wire 2.00 m long must stretch no more than 0.25 cm when a tensile force of 400 N is applied to each end of the wire. What minimum diameter is required for the wire? **1**

   a. 1.9 mm
   b. 1.43 mm
   c. 12.4 mm
   d. 2.48 mm

3. Determine the volume contraction of a solid copper cube, 10 cm on an edge, when subjected to a hydraulic pressure of 7.0 $\times$ 10^6 Pa. Bulk modulus of copper 140 GPa. **1**

   a. 0.05 cm^3
   b. 0.12 cm^3
   c. 0.26 cm^3
   d. 0.08 cm^3
4. A specimen of oil having an initial volume of 600 cm^3 is subjected to a pressure increase of 3.6 $\times$ 10^6 Pa and the volume is found to decrease by 0.45 cm^3 what is the bulk modulus of the material? **1**
   a. 4.4 $\times$ 10^9 Pa
   b. 5.0 $\times$ 10^9 Pa
   c. 4.8 $\times$ 10^9 Pa

d. 4.6×10^9 Pa

5. Elastomers are materials **1**

   a. which cannot be stretched to cause large strains
   b. which can be stretched to cause large strains
   c. which can be stretched without corresponding stress
   d. which cannot be stretched to beyond elastic limit

6. When is a body said to be under compressional strain? **1**

7. The stretching of a coil spring is determined by its shear modulus. Why? **1**

8. An elastic wire is cut to half its original length. How would it affect the maximum load that the wire can support? **1**

9. A particle is thrown upwards. It attains a height (h) after 5 seconds and again after 9s when it comes back. What is the speed of the particle at a height h? **2**

10. A steel cable with a radius of 1.5 cm supports a chairlift at a ski area. If the maximum stress is not to exceed 10^8 N m^{-2}, what is the maximum load the cable can support? **2**

11. Write the characteristics of displacement. **2**

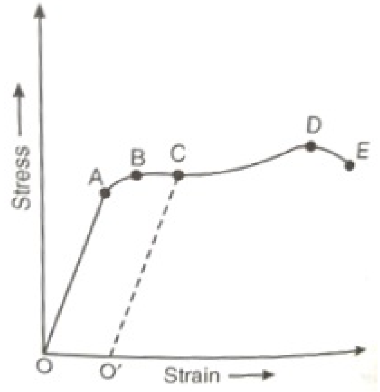
12. The edge of an aluminum cube is 10 cm long. One face of the cube is firmly fixed to a vertical wall. A mass of 100 kg is then attached to the opposite face of the cube. The shear modulus of aluminium is 25 GPa. What is the vertical deflection of this face? **3**

13. The Young's modulus of steel is 2.0×10^{11} N/m^2. If the interatomic spacing for the metal is 2.8×10^{-10} m, find the increase in the interatomic spacing for a force of 10^9 N m^{-2} and the force constant? **3**

14. The stress-strain graph for a metal wire is given in the figure. Up to the point B, the wire returns to its original state O along the curve BAO, when it is gradually unloaded. Point E corresponds to the fracture point of the wire. **3**

   i. Up to which point of the curve, is Hook's law obeyed? This point is also called 'Proportionality limit'.

ii. Which point on the curve corresponds to elastic limit and yield point of the wire?
iii. Indicate the elastic and plastic regions of the stress-strain curve.
iv. What change happens when the wire is loaded up to a stress corresponding to point C on a curve, and then unloaded gradually?



15. Draw a typical stress-strain curve for a ductile metal and briefly explain the important points (salient features) of the curve. **5**